VO NGUYEN DANG KHOA

SENIOR FULL-STACK, SYSTEMS & SECURITY ARCHITECT

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NEXT.JS / GO / RUST
OPEN TO WORK

Full-Stack (Next.js 15 / React 19 / Go) • Rust & Linux Systems • Cryptographic Security & DRM • Self-Hosted Cloud

GRID SYSTEM // ISO-216 A4

/01 EXECUTIVE THESIS

Architecture & Scope

Senior Full-Stack, Systems, and Security Architect with 9+ years in technology and 18+ production applications deployed across Next.js 15 / React 19, Go (Gin), Rust (Axum / Linux kernel), and Docker. Engineered defense-in-depth production systems including Ed25519 offline cryptographic licensing, Argon2id + RFC 6238 TOTP authentication, forensic anti-leak DRM, 42 AI MCP server tools, and dual-architecture static Musl toolchains—backed by enterprise governance across 6 Southeast Asian markets at Procter & Gamble.

18+ Apps	Ed25519	Zero-Leak	<10MB
PRODUCTION ECOSYSTEM	CRYPTOGRAPHIC CONTROLS	FORENSIC ANTI-LEAK DRM	MEMORY-SAFE RUST DAEMONS
Rust, Go, Next.js 15 & DSM 7.2+	Argon2id + TOTP 2FA Sandboxing	Dynamic Watermarking & Blanking	Rootless /dev/uinput & /dev/fb0

01. Full-Stack & Agentic Cloud Systems
Architected kv-synology (Next.js 15, React 19, 42 AI MCP tools) and kv-tube (Go/Gin HLS streaming engine) with multi-container Docker orchestration.

02. Defense-in-Depth & Cryptography
Engineered Argon2id password hashing, RFC 6238 TOTP 2FA, Ed25519-signed offline licensing, forensic DRM (ola), and canonicalized path sandboxing.

03. Rust Kernel & Static Musl Toolchains
Built pure Rust Linux IME (vietc, 60+★) and dual-arch static Musl binaries (kv-trimui) with automated GnuPG package signing and CI/CD release assurance.

/02 CLOUD & WEB SYSTEMS

syno.vndns.net // SPK

kv-synology — Flagship DSM Web Control Plane

[Next.js 15 · React 19 · 42 AI MCP Tools · GnuPG CI/CD] https://syno.vndns.net

Automated SPK repository with GnuPG cryptographic signature verification, dynamic port allocation (20000–59999), QuickConnect relay resolver, and live daemon control (SMB, NFS, SSH, WebDAV, AFP).

ola — Forensic Anti-Leak DRM
Dynamic moving watermark overlay (IP, timestamp, ID) and auto-blank traps upon screen-recording or DevTools inspection.
Go • React • Synology SPK

kv-file-pro — Cryptographic Sandboxing
Argon2id hashing, RFC 6238 TOTP 2FA, Ed25519 asymmetric offline licensing, and dunce::canonicalize path isolation.
Rust (Axum) • SQLite WAL

SysVis.AI — Threat Model Visualizer
Architecture topology and threat boundary mapping for Architecture Review Forums with local-first WebGPU AI inference.
React 19 • WebLLM Qwen3

kv-server — Zero-Trust Tunnel Gateway
Handheld reverse-proxy enforcing Cloudflare Zero-Trust Tunnels, token-based auth (?auth=token), and loopback port isolation.
Rust • Musl • Cloudflared

kv-tube — Hardened Media Engine
Per-IP rate limiting (300 req/min), strict CORS origin allow-lists, and multi-service container network segmentation.
Go (Gin) • Next.js • HLS

spkrepo — Supply Chain Packaging
Automated SPK package distribution platform with GnuPG cryptographic signature verification and container validation.
GnuPG • Docker • Python

/03 RUST & KERNEL SYSTEMS

VietC // Rust 1.85+

VietC (Viet+) — Native High-Performance Linux Input Method Engine

github.com/vndangkhoa/vietc (60★)

Direct Wayland Virtual Keyboard
Direct Unicode keystroke injection via wtype (zwp_virtual_keyboard_v1) and /dev/uinput. Eliminates pre-edit underline flashing and preserves IDE autocompletion in VS Code and GPU terminals.
wtype • /dev/uinput • <1ms Latency

Intelligent Auto-Restore
Dual-layer Vietnamese phonological analysis paired with a technical English lexicon automatically preserves programming keywords (search, test, filter) without manual mode toggling.
Phonology Engine • Context Awareness

Hardware Event Filtering
Strict device path binding to /dev/input/by-path/*-event-kbd eliminates duplicate keystrokes caused by multi-interface 2.4GHz wireless USB dongles. Consumes <10MB RAM as a rootless daemon.
Hardware Binding • <10MB Footprint

/04 WORK EXPERIENCE

2020 — Present

2024 — PRES — Lead Systems & Full-Stack Architect — KV Open-Source & Cloud Ecosystem
+ Architected 18+ production apps across Next.js 15/React 19 (kv-synology + 42 MCP tools), Go/Gin (kv-tube, ola DRM), and Rust (kv-file-pro with Argon2id/TOTP/Ed25519, vietc Linux IME, kv-trimui).

2025 — PRES — AI & Creative Technology Manager — Phibious Vietnam
+ Engineered automated generative AI production pipelines (ComfyUI, FLUX.1 LoRA, Python/CUDA node orchestration) and managed GPU infrastructure ROI, accelerating throughput by 60%.

2020 — 2025 — eCommerce Design & Systems Lead — Procter & Gamble (P&G) SEA
+ Governed regional digital systems, automated QA validation, and modular component architecture across 6 Southeast Asian markets, accelerating delivery 3× across 200+ releases.

/05 SKILLS & CREDENTIALS

Tech Stack & RMIT

FULL-STACK & APIS
TypeScript · React 19 · Next.js 15 · Go (Gin) · Rust (Axum) · Node.js · REST / WebSocket · HLS Video · MCP Tools

SYSTEMS & LOW-LEVEL
Rust 1.85+ · Linux (Arch/Debian) · /dev/uinput & fb0 · Static Musl · POSIX C · Bash · systemd rootless · SQLite WAL

SECURITY & CRYPTOGRAPHY
Argon2id · RFC 6238 TOTP · Ed25519 · GnuPG Signing · Forensic DRM · Sandboxing · Rate Limiting · Zero-Trust

CLOUD & DEVOPS
Docker Multi-Arch · Synology DSM 7.2+ · Nginx / SSL · Cloudflare Tunnels · GitHub Actions · Forgejo CI/CD · Btrfs RAID

Education & Cloud Credentials
RMIT University Vietnam (B.Des. Multimedia Design, 2016) • Google Cloud Certified — Professional Cloud Architect (2023).

Gateway, Storage & CI/CD
Btrfs RAID immutable snapshots, Nginx wildcard SSL reverse proxy, Cloudflare Zero-Trust Tunnels, and GnuPG-signed SPK distribution.

Verified Endpoints
https://syno.vndns.net · github.com/vndangkhoa/vietc · github.com/vndangkhoa/kv-synology

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EMBEDDED SYSTEMS & HANDHELD OS ARCHITECT

trimui.vndns.net
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DUAL-ARCH MUSL
/DEV/FB0 60FPS
6 OS DISTROS
Ho Chi Minh City, VN

Linux Framebuffer (/dev/fb0) • Dual-Arch Static Musl • 6 Handheld OS Distros • MPV & ALSA Hi-Fi

GRID SYSTEM // ISO-216 A4

/01 SYSTEM THESIS

A523 & A133P OS

Creator and lead systems architect of KV-TrimUI (trimui.vndns.net)—the complete open-source native software ecosystem and 18-application suite for TrimUI handheld consoles (Smart Pro S TG5050 Allwinner A523 Octa-Core, Smart Pro TG5040 A133P, Brick Pro/Brick). Engineered a Dual-Engine Architecture combining a 100% pure Rust 60 FPS direct framebuffer engine (/dev/fb0 mmap, <900KB–5MB RAM) with lightweight Next.js 15 + dual-arch static Musl daemons (aarch64 & armv7l, <15MB RAM), turnkey packagers across all 6 major OS distributions (Stock OS, SpruceOS, Knulli, NextUI, NXRedux, GammaOS), and a 1-Click Web Installer & dual-mirror OTA distribution network.

18 Apps	6 OS Distros	< 900 KB	Dual-Arch
PRODUCTION HANDHELD SUITE	**MULTI-OS TURNKEY SUPPORT**	**PURE RUST 60 FPS ENGINE**	**STATIC MUSL TOOLCHAINS**
Tube, Tidal, Netflix, TV, Radio, Play, Chat	Stock, Spruce, Knulli, NextUI, NXRedux, Gamma	Direct /dev/fb0 mmap • Zero X11/Wayland	aarch64 & armv7l • A523 & A133P SoCs

/02 KERNEL PILLARS

Dual-Engine · 6-OS

SUBSYSTEM 01

Dual-Engine Architecture (Pure Rust FB + Web Hybrid)

Engine B renders 100% native Rust directly to /dev/fb0 via mmap at 60 FPS (<900KB–5MB RAM) with DRM Plane 93 fail-safe; Engine A pairs Next.js 15 static UIs with static Musl daemons (<15MB RAM) and @kv-trimui/shared-gamepad OSK.

SUBSYSTEM 02

Dual-Arch Static Musl & 6-OS Turnkey Packagers

Cross-compiles zero-dependency static binaries for both aarch64-unknown-linux-musl and armv7-unknown-linux-musleabihf to bypass GLIBC 2.28 limits across Stock OS, SpruceOS, Knulli, NextUI, NXRedux, and GammaOS.

SUBSYSTEM 03

Hardware MPV 720p60 & Bit-Perfect ALSA Hi-Fi

Streams YouTube, cinema, and IPTV via GPU-accelerated MPV (--vo=gpu, Cedar/Mali) and routes Tidal Hi-Fi, Apple Music & AirPlay directly to ALSA hw:0,0 with A523 DACL unity gain (160) tuning.

/03 HANDHELD SUITE

18 Production Apps

kv-radio [PURE RUST 60 FPS FRAMEBUFFER · <900KB RAM]

Rust 1.85 · /dev/fb0 mmap · Direct ALSA

Standalone embedded internet radio station player directly rendering to the Linux framebuffer with double-buffered page flipping, custom TrueType glyph rasterization, dirty-rectangle blitting, and direct ALSA PCM streaming.

kv-tube & kv-tv — YOUTUBE & LIVE IPTV
Ad-free InnerTube YouTube client + 24/7 Live IPTV (VTV, HTV, NHK, Bloomberg) with 720p60 MPV GPU playback.
Next.js 15 • Rust FB • MPV HWDEC

kv-tidal & kv-airplay — HI-FI & AIRPLAY AUDIO
Bit-perfect lossless streamer (Tidal, Apple Music, NAS Vault) + Shairport Sync wireless AirPlay receiver via ALSA hw:0,0.
Rust Musl • Direct ALSA • DACL 160

kv-netflix & kv-stremio — HANDHELD CINEMA
On-demand movie & anime streaming clients with 6-digit phone pairing, episode selectors, and bilingual EN/VI subtitles.
Native Rust /dev/fb0 • HLS • MPV

kv-play & kv-theme — CLOUD ROMS & STYLER
Ozone-styled cloud ROM store with auto .7z/.cue extraction & self-healing cores + real-time handheld theme/icon switcher.
100% Native Rust /dev/fb0 • Multi-OS

kv-news & kv-anime — READER & MANGA 60FPS
Kindle-style ad-free news reader (VnExpress, Tuổi Trẻ) and NetTruyen manga reader with aspect scaling & D-Pad paging.
100% Native Rust Framebuffer

kv-chat, server & torch — AI, TUNNEL & UTILITY
Voice AI companion (Safari WebGPU/LLMs), Cloudflare Tunnel web server daemon, and QR phone-paired LED marquee strobe.
Rust • Cloudflared • WebGPU • WS

/04 FIRMWARE LOG

2026 Roadmap

2025 — PRESENT
KV-TrimUI 18-App Ecosystem & Multi-OS Packaging Pipeline

System Architect & Creator

+ Architected 18 production apps (kv-tube, kv-tidal, kv-netflix, kv-radio, kv-tv, kv-news, kv-anime, kv-stremio, kv-play, kv-theme, kv-server, kv-airplay, kv-chat, kv-torch) for Smart Pro S (A523), Smart Pro (A133P), and Brick.
+ Built automated multi-OS packaging (pnpm package:stock/spruce/knulli/nextui/nxredux & cook_gamma.sh) plus cook.sh single-source dual-mirror OTA release workflow (GitHub + MyDS Forgejo).

2025 — PRESENT
Native Rust Framebuffer Engine, Shared Gamepad & AI Knowledge Hub

Core Engine & Localization Lead

+ Engineered <900KB pure Rust /dev/fb0 engine (kv-radio), compile-time bilingual EN/VI localization (kv-i18n & shared-i18n), and @kv-trimui/shared-gamepad with instant X-button On-Screen Keyboard.
+ Published machine-readable AI Agent Knowledge Hub (llms-full.txt, /api/v1/agent/knowledge, and trimui-rust-starter template) documenting DRM Plane 93 fail-safes and Evdev polling.

/05 EMBEDDED SPEC

Matrix & OTA Links

HARDWARE & SOC TARGETS
Allwinner A523 Octa-Core · Allwinner A133P · Smart Pro S (TG5050) · Brick Pro / Brick · DRM Plane 93 Fix · PWM Fan & Hall Sticks

DUAL-ENGINE & COMPILERS
Rust 1.85+ (/dev/fb0 mmap) · aarch64 & armv7l Musl · Next.js 15 Static · @kv-trimui/shared-gamepad · kv-i18n (EN/VI) · TrueType Rasterizer

MEDIA & KERNEL SUBSYSTEMS
MPV GPU (--hwdec=auto) · ALSA Direct PCM (hw:0,0) · A523 DACL Unity Gain 160 · Linux Evdev Polling · Shairport AirPlay · Safari WebGPU LLM

6-OS PACKAGING & OTA
Stock OS · SpruceOS · Knulli (Batocera) · NextUI / MinUI (.pakz) · NXRedux OS · GammaOS (Android 13) · cook.sh Dual-Mirror OTA

https://trimui.vndns.net
Official App Store & 1-Click Web Installer (/installer)

github.com/vndangkhoa/kv-trimui
Primary Release & Dual-Mirror MyDS Forgejo Repo

trimui.vndns.net/llms-full.txt
Developer & AI Agent Framebuffer Knowledge Hub